

Bimukti Mozzumdar

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U.S. Permanent Resident (Green Card) · No sponsorship required

EDUCATION

CUNY Hunter College

New York, NY

Bachelor of Arts in Computer Science; GPA: 3.94/4.0

Expected Jan. 2028

- **Coursework:** Data Structures & Algorithms, Comp. Architecture, Software Engineering, Computation Theory, Web Dev.

EXPERIENCE

Machine Learning Research Assistant

Mar. 2026 – Present

CUNY Hunter College

New York, NY

- Design a model for synthetic RNA-seq data conditioned on tissue type, addressing the scarcity of labeled expression samples available to cancer researchers.
- Compress 20,000+ genomic features into discrete latent codes with a VAE, using Gumbel-Softmax so the discrete sampling stays differentiable end-to-end.

Software Engineering Intern

Jun. 2025 – Aug. 2025

Unadat

New York, NY

- Cut weekly manual data entry 90% by writing a Python pipeline that parsed school-sent emails into MySQL schema.
- Enabled case managers to catch student absence earlier by shipping threshold-triggered alerts to the internal dashboard.
- Removed manual contract signing by integrating the DocuSign API with a Node.js backend to automate authentication.

Software Engineering (Google-SWEP) Fellow

Mar. 2026 – Present

Google

New York, NY

- Selected as 1 of 250 from 1,900+ applicants for a year-long technical development program.
- Sit weekly mock interviews with Google engineers and drill data structures and algorithms with a Google-assigned mentor.

Computing Lab & IT Assistant

Oct. 2024 – Present

Social Sciences Computing Lab, CUNY Hunter College

New York, NY

- Support 300+ students weekly across 43 machines: OS imaging, troubleshooting, research-software upkeep.
- Lead 10 student workers, owning scheduling, onboarding, and escalation for lab-wide incidents.

PROJECTS

Redis Clone

Apr. 2026 – Present

Go, TCP, Hash Tables, Concurrency

[GitHub](#)

- Built an in-memory key-value store in Go on a custom hash table (MurmurHash3, separate chaining).
- Implemented progressive rehashing, migrating a few buckets per operation to avoid O(n) resize stalls.
- Served SET/GET/DEL/KEYS and TTL commands over a length-prefixed binary protocol, one goroutine per connection.
- Benchmarked against sync.Map and map+RWMutex: within 20 ns/op on single-threaded get/set despite carrying progressive-rehash overhead.

DegreeFlow

May 2026 – Present

React, TypeScript, Node.js, Prisma, PostgreSQL

[GitHub](#) | [Live Demo](#)

- Led both backend & frontend of an automated degree-audit tool built by a four-person team.
- Developed a recommender that generates future course plans across an 8,800+ course catalog.
- Ranked top 5 instructors for each recommended course by a Bayesian-weighted rating over 72,518 scraped student reviews.

PeerDrop

Oct. 2025

Go, WebRTC, JavaScript, Docker

[GitHub](#)

- Built a browser based file transfer: peers pair by short code, then send over a WebRTC data channel.
- Limited the Go WebSocket server to signaling, so file data never passes through or rests on the server.
- Ran on 43 lab machines for two semesters, replacing cloud-storage logins on shared computers.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, C++, SQL

Frameworks & Libraries: React, Node.js, Express, Django, PyTorch, Hugging Face Diffusers, Pandas, NumPy, scikit-learn

Databases: PostgreSQL, MySQL, Redis, Prisma (ORM)

Tools & Platforms: Git, Docker, Linux, AWS S3